

LRF-to-TRF Automated Conversion Pipeline

Stage 1

Parsing and Deny-All Normalization



Stage 2

Anomaly Detection (Deterministic Triad)
exhaustive $O(n^2)$ · exact integer arithmetic



Stage 3

Attribute Ordering Selection (1 of 12)



Stage 4

Unified 4D Range Decomposition



Stage 5

Projection Normalization (Algorithm 1)



Stage 6

TRF Construction